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Jaemin Kim

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Research Interest

Aiming to improve the controllability of advanced generative models to **maximize rewards**. Recently contributed to this goal by exploring efficient guidance methods, such as training-free [C2,C4,C5], Plug-and-Play [C3, C6], and RL-based agentic search [W1, C8].

Education

Korea Advanced Institute of Science and Technology (KAIST) — Mar 2025 - Now
Ph.D. in Artificial Intelligence — *Advisor: Prof. Jong Chul Ye*

Korea Advanced Institute of Science and Technology (KAIST) — Mar 2023 - Feb 2025
M.S. in Artificial Intelligence — *Advisor: Prof. Jong Chul Ye*

Korea Advanced Institute of Science and Technology (KAIST) — Mar 2016 - Feb 2023
B.S. in Bio and Brain Engineering, Electrical Engineering (Double Major) — *GPA: 3.96/4.3 (Magna Cum Laude)*

Publications *: Equal Contribution

[C8] **Dementia-R1: Reinforced Pretraining and Reasoning from Unstructured Clinical Notes for Real-World Dementia Prognosis** — *EMNLP 2026*
Choonghan Kim*, Hyunmin Hwang*, Hangeol Chang*, **Jaemin Kim***, Jinse Park, Jae-Sung Lim, Jong Chul Ye

[C7] **Adaptive Guidance for Retrieval-Augmented Masked Diffusion Models** — *EMNLP Findings 2026*
Jaemin Kim and Jong Chul Ye

[C6] **Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs** — *ICML 2026*
Jaemin Kim*, Hangeol Chang*, Hyunmin Hwang*, Choonghan Kim, Jong Chul Ye

[C5] **Training-Free Reward-Guided Image Editing via Trajectory Optimal Control** — *ICLR 2026*
Jinho Chang*, **Jaemin Kim***, Jong Chul Ye

[C4] **Free^{2}Guide: Training-Free Text-to-Video Alignment using Image LVLM** — *ICCV 2025*
Jaemin Kim, Bryan Sangwoo Kim, Jong Chul Ye

[C3] **Optical-Flow Guided Prompt Optimization for Coherent Video Generation** — *CVPR 2025*
Hyelin Nam*, **Jaemin Kim***, Dohun Lee, Jong Chul Ye

[C2] **Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI** — *CVPR 2025*
Won Jun Kim*, Hyungjin Chung*, **Jaemin Kim***, Sangmin Lee, Byeongsu Sim, Jong Chul Ye

[C1] **Generalized Consistency Trajectory Models for Image Manipulation** — *ICLR 2025*
Beomsu Kim*, **Jaemin Kim***, Jeongsol Kim, Jong Chul Ye

Pre-prints and Workshops *: Equal Contribution

[W1] **AgentPSO: Evolving Agent Reasoning Skill via Multi-agent Particle Swarm Optimization** *ICML AI4Math 2026*
Hyunmin Hwang*, **Jaemin Kim***, Choonghan Kim, Hangeol Chang, Jong Chul Ye

[P2] **DMD-augmented Unpaired Neural Schrödinger Bridge for Ultra-Low Field MRI Enhancement** *Preprint 2026*
Youngmin Kim*, Jaeyun Shin*, Jeongchan Kim*, Taehoon Lee, **Jaemin Kim**, Peter Hsu, Jelle Veraart, Jong Chul Ye

[P1] **HiCBridge: Resolution Enhancement of Hi-C Data Using Direct Diffusion Bridge** *Preprint 2024*
Jaemin Kim and Jong Chul Ye

Experiences

Research Intern May 2026 - Oct 2026
Sony Group - Creative AI Lab *Tokyo Osaki, Japan*

Karlsruhe Institute of Technology (KIT) Sep 2019 - Feb 2020
Electrical Engineering and Information Technology (Exchange Student) *Karlsruhe, Germany*

Patents

Method and Apparatus for Approximating Gradient of Artificial Neural Network Model 2025
- (South Korea) Patent No.10-2025-0145337

A Single, Composable Plug-and-play Reasoner for Frozen LLMs 2025
- (South Korea) Patent No.10-2025-0132734

Honors and Awards

32nd Samsung Humantech Paper Award Silver Prize
Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs *Track: Signal Processing*

Dongwon-KAIST scholarship Mar 2023 - Feb 2025
Full scholarship for tuition and stipend

Baden-Württemberg Stipendium Sep 2019 - Feb 2020
Scholarship for international students to study at a university in Baden-Württemberg

South Korea National Science & Technology Scholarship Mar 2018 - Mar 2020
Scholarship for attracting outstanding talents

Academic Activities

Reviewer
ICLR 2026, ICML 2026, NeurIPS 2026, TMLR

Persident, Student Council May 2025 - Mar 2026
KAIST AI

Teaching Assistant Mar 2025 - Jun 2025
KAIST - AI618: Generative Models and Unsupervised Learning

References

Cristian Simon Mentor at Sony
u6562565@anu.edu.au *AI Researcher, Creative AI, Sony Group*

Jong Chul Ye M.S., Ph.D. Advisor
jong.ye@kaist.ac.kr *Endowed Chair Professor, Kim Jaechul Graduate School of Artificial Intelligence, KAIST*